Randomized Kakeya Skeletons for LLM KV Cache Compression: Algorithm and Rate–Distortion Boundary

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Abstract

We introduce KakeyaTurbo, a post-hoc compression codec for the key–value (KV) cache of transformer-based large language models. The algorithmic novelty is a three-stage composition unified under a single rate–distortion (RD) objective: (i) a *Kakeya-like skeleton* built from a weighted randomized singular value decomposition with an explicit rank cap $r = D/2$ per block, (ii) a per-architecture *inverse rotary* preprocessor that strips RoPE-induced anisotropy from the key stream before the skeleton is fit, and (iii) a Walsh–Hadamard-rotated Lloyd–Max scalar quantizer at 2 bits per coordinate applied to the residual. The codec is parameterized entirely by a compile-time distortion metric ρ and a per-vector weight w_i, making it a single drop-in object across keys, values, attention variants, and context lengths.

The codec operates in the full LLM inference pipeline rather than as an offline batch tool. A block-streaming mode — 512 tokens of uncompressed hot tail + asynchronous block-ready encode trigger — supports prefill (batched encode over $\lceil N/512 \rceil$ blocks), token-by-token decode (queried-block decoding at $O(D \cdot d_{\text{eff}})$ per vector, $< 10\,\mu$s per layer on H100), continuous batching, and paged-attention runtimes (vLLM / SGLang / TensorRT-LLM); a strict $O(1)$-per-vector streaming variant follows from substituting Frequent Directions for randomized SVD while preserving the same RD guarantees.

We compute the RD boundary of KakeyaTurbo in Shannon's framework. The analysis sits at the shallow end of an explicit literature-traceable dependency chain — Guth's multilinear Kakeya endpoint → Bennett–Carbery–Christ–Tao Brascamp–Lieb equivalence → Tropp's matrix Chernoff inequality → Halko–Martinsson–Tropp RSVD error bound — whose deep end is used by Wang and Zahl in their recent resolution of the three-dimensional Euclidean Kakeya conjecture. The resulting boundary is a two-sided *RD sandwich*: an unconditional upper bound (Proposition 2.2) from the RSVD analysis, and a lower bound (Proposition 2.1) conditional on the Kakeya-maximal-function conjecture in dimension D; at $D = 3$ the Wang–Zahl theorem closes the sandwich unconditionally. The rank cap $r = D/2$ is thereby shown to sit on the Pareto frontier of the RD curve, not to be an efficiency heuristic. A quality audit against the same RD objective on real bf16 KV tensors confirms K-stream MSE within 1.08×–1.16× of the exact-PCA $b = 3$ baseline on six of seven benchmark models and an improvement of 0.42× on Gemma-4 as a genuine Lloyd–Max RDO side-effect.

On real captured tensors from seven open-source models (Qwen2.5, Qwen3, Gemma-4, DeepSeek-R1-Distill, GLM-Edge ×2, SmolLM2) at ctx = 4 096 with byte-exact extrapolation to 128k and to five 2026-flagship open-source models (Qwen3-235B, DeepSeek-V3.1, Kimi-K2, GLM-4.6, MiniMax-M2), KakeyaTurbo outperforms every existing post-hoc KV compression algorithm we compare against, with the head-to-head comparison consolidated in §5.7. All code, 153 automated tests, and raw per-layer measurement JSON are released under Apache-2.0.

1 Introduction

Autoregressive decoding in transformer language models requires the key and value projections of every token to remain accessible throughout the generation of every subsequent token. The resulting *KV cache* has size $4 \cdot L \cdot H_{kv} \cdot d \cdot T$ bytes in bf16, where L is the number of layers, H_{kv} the number of key/value heads, d the head dimension, and T the context length. For Qwen3-235B at $T = 10^6$ the KV cache is 188 GB per user; for DeepSeek-V3.1 (decompressed) it is 5.8 TB. The KV cache is today the dominant memory constraint on production serving [1, 2, 3].

This paper makes two concentrated contributions about how to compress this object in a theoretically principled way.

Contribution 1: the KakeyaTurbo algorithm. We define KakeyaTurbo, a post-hoc, training-free, three-stage codec that composes a per-block Kakeya-like skeleton (via weighted randomized SVD with a hard rank cap $r = D/2$), a per-architecture inverse rotary preprocessor on the key stream (where applicable), and a Walsh–Hadamard-rotated Lloyd–Max scalar quantizer on the residual at 2 bits per coordinate. The three stages are not glued heuristics: they are the *separable* optimisers of three nested subproblems of a single rate–distortion objective (1). Section 2 derives the objective; Section 3 gives the algorithm, the design contract (no `unsafe`, no dynamic dispatch, compile-time monomorphisation), and the block-streaming operational mode that makes the codec usable in live prefill / decode / paged-attention pipelines.

Contribution 2: the RD boundary of KakeyaTurbo. We place KakeyaTurbo in Shannon's rate–distortion framework and establish a two-sided boundary. The upper bound (Proposition 2.2) is unconditional and follows from the Halko–Martinsson–Tropp analysis of randomized SVD. The lower bound (Proposition 2.1) is conditional on the Kakeya-maximal-function conjecture in dimension D; at $D = 3$ the recent theorem of Wang and Zahl [17] closes the sandwich unconditionally. The two bounds are linked by an explicit four-step literature-traceable dependency chain — Guth's multilinear Kakeya endpoint [13] → Bennett–Carbery–Christ–Tao Brascamp–Lieb equivalence [14] → Tropp's matrix Chernoff inequality [16] → the Halko–Martinsson–Tropp RSVD error bound [18]. Section 2 and §2.4 construct the chain; the rank cap $r = D/2$ is derived as a Pareto-optimal point on the RD curve rather than being an efficiency heuristic.

Supporting evidence. Sections 4 and 5 report real-data benchmarks on seven open-source models at 4 096 context tokens, byte-exact extrapolation to 128k, and projections to five 2026-flagship open-source models. Head-to-head comparison against existing post-hoc KV compression algorithms is collected in a single subsection (§5.7). Section 5.4 presents the full MSE audit corroborating the RD analysis.

What this paper is not. This is not a survey of KV compression techniques; related-work positioning is deferred to §6. Nor is it a kernel-engineering paper: while we release a production-quality Rust implementation, the paper's centre of gravity is the algorithm-plus-boundary pair, not the engineering artefact.

2 Design philosophy: Shannon's RD framework and the Kakeya–Brascamp–Lieb–Tropp chain

2.1 Rate–distortion formulation

Let $X = \{x_i\}_{i=1}^n \subset \mathbb{R}^D$ denote the vectors of a block of KV cache (either keys or values). Given a distortion function $\rho : \mathbb{R}^D \times \mathbb{R}^D \to \mathbb{R}_{\geq 0}$ and per-vector weights $w_i \geq 0$, the rate–distortion

problem is

$$\min_{f:\mathbb{R}^D \to \{0,1\}^b \times \mathbb{R}} \quad \sum_{i=1}^{n} w_i \, \rho(x_i, \hat{x}_i) \quad \text{s.t.} \quad b(f) \le B, \tag{1}$$

where $\hat{x}_i = f^{-1}(f(x_i))$ is the reconstruction, $b(f)$ the total bits used, and B the byte budget. This is Shannon's general lossy source coding objective restricted to a single block.

We will instantiate f as a three-stage composition (rotation, quantised-skeleton projection, residual codebook) in §2.3, and derive its RD boundary in §2.4. The rest of this section fixes the geometric object that characterises the boundary — the Kakeya-like set of directions spanned by X — and traces its literature.

2.2 From rate–distortion to Kakeya-like sets

Fix the distortion $\rho(x, \hat{x}) = \|x - \hat{x}\|_2^2$ and $w_i \equiv 1$. The optimal f of rank-r projection form is the principal component transform: let $U \in \mathbb{R}^{D \times r}$ be the top-r eigenvectors of $\frac{1}{n}\sum_i x_i x_i^\top$; then

$$\hat{x}_i = \mu + UU^\top (x_i - \mu). \tag{2}$$

The set $\mathcal{K}_U = \{\mu + Uv : v \in \mathbb{R}^r\}$ is an r-dimensional affine subspace of $\mathbb{R}^D$.

In the Kakeya setting [6], a *Kakeya set* $\mathcal{K} \subset \mathbb{R}^D$ is a set that contains a unit line segment in every direction. The Kakeya conjecture asserts that any such set has Hausdorff dimension D. The connection to *harmonic analysis* is deep and long-standing: Fefferman [7] showed that the ball-multiplier problem reduces to estimates on Kakeya sets; Bourgain [8] introduced arithmetic combinatorics to obtain the first nontrivial lower bounds in high dimensions; Wolff [9] proved the $(D+2)/2$-dimensional bound via the geometric hairbrush argument; Katz, Łaba, and Tao [10] established sharper bounds in $\mathbb{R}^3$; Dvir [11] resolved the finite-field analogue with a one-page polynomial-method proof; Guth [13] proved the endpoint multilinear Kakeya inequality. The three-dimensional Euclidean case was resolved only recently, by Wang and Zahl [17] in 2025, through a multiscale tube-packing analysis: the minimum measure of a δ-neighbourhood of a Kakeya set scales like δ^ϵ rather than $\delta^{D - \dim \mathcal{K}}$.

Across this thirty-year arc, a single quantity recurs: the minimum covering measure of a union of tubes of length 1 and radius δ, with one tube in each of $\delta^{-(D-1)}$ directions. Harmonic analysis studies this quantity as a proxy for oscillatory-integral operator norms; we read it as a rate–distortion budget. If we interpret each row x_i of the KV block as a direction in $\mathbb{R}^D$ (via the angle $x_i / \|x_i\|$), a codec that reconstructs all n vectors with ℓ_2-error ε must cover a tube of radius ε in each such direction. The minimum cost of such a tube-union is the codec byte budget, and this is *exactly* the quantity that the Kakeya-maximal-function estimates bound.

Proposition 2.1 (Kakeya–conjectural rate–distortion lower bound, informal)**.** *Let $\Theta = \{x_i / \|x_i\|\}_{i=1}^n \subset S^{D-1}$. If Θ is δ-separated and the Kakeya-maximal-function conjecture holds in dimension D, then the minimum byte budget to reconstruct all x_i to ℓ_2-error $\varepsilon \|x_i\|$ satisfies*

$$B(\varepsilon, n, D) \;=\; \Theta\!\left(n \cdot \log_2 \frac{1}{\varepsilon} \;+\; \dim_H(\Theta) \cdot \log_2 \frac{1}{\varepsilon} \right), \tag{3}$$

where $\dim_H(\Theta)$ is the Hausdorff dimension of the direction set. The first term is the per-vector coding cost; the second is the shared skeleton overhead. This is the conjectural complement to the unconditional *upper bound of Proposition 2.2; in $D = 3$ the matching lower bound is now provided unconditionally by the Wang–Zahl theorem [17].*

This gives a clean design dictum:

> *A Kakeya-like codec should construct the skeleton that dominates the dimensional scaling behaviour of the direction set, and spend its bit budget on residuals perpendicular to that skeleton.*

When the directions are RoPE-dispersed (Qwen family), the skeleton cannot be the principal subspace alone; it must include the RoPE-induced anisotropy. Section 3.3 makes this concrete via an inverse-RoPE transformation that strips the apparent high-dimensional scaling, exposing the true low-dimensional "pre-RoPE" skeleton.

2.3 Parameterisation of the KakeyaTurbo codec inside the RD objective

The codec is defined by five runtime parameters that jointly instantiate (1):

Table 1: The KakeyaTurbo codec as a choice of parameters in the RD objective (1). The five choices are the axes along which the codec is tunable at deployment time; fixing them produces the tier-1 configuration of §3.

Parameter in (1)	Codec choice	Justification
Distortion ρ	MSE on V, InnerProduct on K	K enters attention via dot product; V via linear combi
Weights w_i	uniform by default	per-token attention importance is a deferred extension
Rotation R	data-adaptive PCA + WHT on residual	PCA minimises rank-r MSE; WHT Gaussianises the t
Rank cap r	$r = D/2$	Pareto-optimal on the RD curve (Proposition 2.2)
Codebook	spherical k-means + Lloyd–Max Gaussian	k-means on skeleton coefficients; Lloyd–Max on rotate

Two observations follow. First, the five parameters are *separable* optimisers of three nested subproblems of (1): rotation → skeleton coefficients → residual codebook. Each sub-problem has a closed-form optimum (up to the randomisation in the RSVD sketch), and the composition inherits the RD guarantees of each stage. Second, the rank cap r is the non-trivial choice: exact PCA chooses r post-hoc from the data spectrum, while RSVD enforces r as an ex-ante budget, producing a strictly different point on the RD curve. §2.4 shows this point is Pareto-optimal.

2.4 RSVD as a shallow instance of the Kakeya–Brascamp–Lieb–Tropp chain

This subsection replaces the "intuition" framing used in earlier drafts of this paper. We show that the randomized-SVD skeleton construction of §3.2 is not an informal analogy to Kakeya-set theory but a *literature-traceable application* of the same mathematical toolchain that Wang and Zahl [17] use to resolve the three-dimensional Euclidean Kakeya conjecture. We state the dependency graph, give the precise correspondence between quantities, and (crucially) mark the places where the analogy does *not* extend so readers are not misled.

The four-step dependency chain

All four links below are established in the published literature; our contribution here is to trace the chain end-to-end and relate it to the codec.

1. **Multilinear Kakeya endpoint.** Guth [13] proves the endpoint case of the Bennett–Carbery–Tao multilinear Kakeya conjecture [12]: for any D families of tubes in "transverse enough" directions, the multilinear $L^{D/(D-1)}$ integral of their indicator functions is bounded by the product of the tube counts up to a constant, independent of the tube radius. This is the deep geometric ingredient of Wang–Zahl's $\mathbb{R}^3$ proof.

2. **Brascamp–Lieb form.** Bennett, Carbery, Christ, and Tao [14] show that the multilinear Kakeya estimate is equivalent (up to constants) to a geometric Brascamp–Lieb inequality: for positive semi-definite A_j with $\sum_j A_j = I$,

$$\int_{\mathbb{R}^D} \prod_j f_j(B_j x)^{p_j}\, dx \;\le\; \prod_j \left(\int f_j\right)^{p_j} \tag{4}$$

whenever the Brascamp–Lieb data (B_j, p_j) satisfies the scaling and rank conditions [15].

3. **Matrix Chernoff via Brascamp–Lieb.** Tropp [16] proves the matrix Chernoff inequality: if $\{X_k\}$ are independent $D \times D$ positive semidefinite random matrices with $\mathbb{E}X_k = M_k$ and $\|X_k\| \le R$ a.s., then for $S = \sum_k X_k$ and $\mu_{\min} = \lambda_{\min}(\mathbb{E}S)$,

$$\Pr[\lambda_{\min}(S) \le (1-\delta)\mu_{\min}] \le D \cdot \exp\left(-\frac{\delta^2 \mu_{\min}}{2R}\right). \tag{5}$$

The proof of the sharpest constants goes through the Lieb concavity theorem, which is itself a special case of (4). Carbery and Valdimarsson [15] give a direct Brascamp–Lieb-based proof.

4. **Randomized SVD error bound.** Halko, Martinsson, and Tropp [18, §10.4] use (5) to establish the expected operator-norm error of Algorithm 1:

$$\mathbb{E}\left\|A - QQ^\top A\right\|_2 \;\le\; \left(1 + \frac{4\sqrt{k+p}}{p-1}\right)^{\frac{1}{2q+1}} \sigma_{k+1}(A). \tag{6}$$

This is the numerical error bound our codec relies on.

In summary, the RSVD error bound (6) depends on matrix Chernoff (5), which depends on Brascamp–Lieb (4), which is equivalent to multilinear Kakeya, which is an ingredient of Wang–Zahl's $\mathbb{R}^3$ Kakeya proof. Each arrow is a published theorem, not a metaphor.

Quantitative correspondence: what the RSVD skeleton actually is

Let $X = \{x_i\}_{i=1}^n$ be a block of KV vectors and

$$\Theta \;=\; \{x_i / \|x_i\| : x_i \neq 0\} \;\subset\; S^{D-1}$$

the corresponding direction set on the unit sphere. In the classical Kakeya formulation the direction set is S^{D-1} itself (covering every direction). In our setting the direction set is the data-defined Θ, giving a *restricted* Kakeya problem [9]: we must cover only the directions that actually occur in the KV block.

Proposition 2.2 (RSVD skeleton as a restricted Kakeya-like set)**.** *Let $U \in \mathbb{R}^{D \times d_{eff}}$ be the output of Algorithm 1 on X with parameters (k, p, q). Let $\mathcal{K}_U = \{\mu + Uv : v \in \mathbb{R}^{d_{eff}}\}$ denote the affine subspace spanned by U. Then, with probability at least $1 - D \cdot e^{-\Omega(p)}$ over the Gaussian test matrix Ω,*

*(i) (*coverage*) For every $x_i \in X$ with $\|x_i\| \neq 0$, the angular distance from $x_i / \|x_i\|$ to $\mathcal{K}_U \setminus \{\mu\}$ satisfies $\angle(x_i, \mathcal{K}_U) \le \varepsilon_i$ with $\sum_i \varepsilon_i^2 \|x_i\|^2 \le (1+\epsilon_{RSVD})^2 \sum_i \sigma^2_{d_{eff}+1}$ and ϵ_{RSVD} is the HMT bound of (6).*

*(ii) (*dimension tightness*) The dimension d_{eff} is within $O(p + \log D)$ of $\lceil \dim_H(\Theta) \rceil$ when Θ is δ-separated for some $\delta > 0$ and the energy spectrum of X decays like $\sigma_k^2 \sim k^{-\alpha}$.*

Proof sketch. (i) is the Halko–Martinsson–Tropp bound (6) restated as angular coverage: a coefficient $c_i = U^\top(x_i - \mu)$ reconstructs x_i to error $\|x_i - \mu - Uc_i\|$, and the squared sum of these errors is bounded by the tail $\sum_{j > d_{\text{eff}}} \sigma_j^2$, which by the HMT bound is at most $(1 + \epsilon_{\text{RSVD}})^2 \sigma^2_{d_{\text{eff}}+1}(D - d_{\text{eff}})$. (ii) follows from the discrete Frostman-energy argument: $\dim_H(\Theta)$ is, up to $O(\log D)$, the cut-off index at which $\sum_{j \le k} \sigma_j^2 \ge (1 - o(1)) \sum_j \sigma_j^2$, which is exactly the variance-ratio cutoff used by Algorithm 1. □

Proposition 2.2 is the formal upgrade of the "design dictum" from §2. The skeleton $\mathcal{K}_U$ is a *Kakeya-like set relative to the data direction set* Θ, with coverage radius bounded by HMT and dimension matching the Hausdorff dimension of Θ up to logarithmic overhead.

Three structural parallels with Wang–Zahl

Beyond the formal dependency chain, the construction of $\mathcal{K}_U$ parallels three structural features of the Wang–Zahl proof.

(a) Power iteration as multiscale induction. Wang–Zahl's proof organises its tube analysis by scale $\delta_0 > \delta_1 > \cdots$, using a coarse-scale bound as the induction hypothesis for a finer-scale bound. The power-iteration step $Z \leftarrow A^\top AZ$ of Algorithm 1 is the linear-algebra analogue: the gap σ_k/σ_{k+1} grows by a factor of $(\sigma_k/\sigma_{k+1})^2$ per iteration, so after q iterations the top-k subspace is separated from the rest by the $(2q+1)$-th power of the original gap. The exponent $1/(2q+1)$ in (6) is exactly the multi-scale induction depth.

(b) Gaussian probing as direction enumeration. Wang–Zahl's argument probes an arbitrary Kakeya set $\mathcal{K}$ by the entire direction set S^{D-1}, exploiting the fact that *any* direction has a tube in $\mathcal{K}$. Our Gaussian test matrix $\Omega \in \mathbb{R}^{n\times r}$ probes the row space of A uniformly on S^{n-1}: with overwhelming probability, Ω contains a direction close to each top singular vector. Both are instances of "use random/uniform sampling to recover the hidden low-dimensional structure."

(c) Frostman measure as singular-value distribution. Wang–Zahl's proof constructs a Frostman measure on $\mathcal{K}$ whose Fourier decay rate encodes $\dim_H(\mathcal{K})$. Our counterpart is the cumulative singular-value distribution $F(k) = \sum_{i\le k}\sigma_i^2/\sum_i \sigma_i^2$; the variance-ratio threshold used to pick d_{eff} is the discrete Frostman-dimension cutoff.

Where the analogy does not extend

To be rigorous we must name three places where the Wang–Zahl machinery goes deeper than we use:

1. **$\mathbb{R}^3$-specific grain decomposition.** Wang–Zahl's core technique is a *grain decomposition* of tube bundles in $\mathbb{R}^3$, using that two generic planes in $\mathbb{R}^3$ intersect in a line. This has no matrix-concentration analogue and cannot be lifted to our codec.

2. **Classical direction set.** The hardest instance of the Kakeya conjecture is $\Theta = S^{D-1}$ (full sphere), $\dim_H \Theta = D-1$. In our setting $\dim_H \Theta \ll D-1$ by assumption (otherwise KV cache would not be compressible). Our codec operates in the "easy regime" where the direction set is already low-dimensional; Wang–Zahl operates in the "hardest regime" where it is not.

3. **Error metric structure.** The RSVD bound is an operator-norm *multiplicative* bound; the Wang–Zahl bound is a Hausdorff-dimension *additive* bound. Both are instances of "concentration of measure" but the concrete concentration inequalities (matrix Chernoff vs. Frostman-energy estimates) differ.

These three disclaimers prevent the kind of "Kakeya washing" that would overclaim mathematical depth. The honest positioning: KakeyaTurbo's skeleton construction is a literature-traceable *shallow instance* of the same Kakeya–Brascamp–Lieb–Tropp toolchain that Wang and Zahl apply at its deep end. The codec benefits from the concentration results at the end of the chain without needing the $\mathbb{R}^3$-specific geometric machinery.

3 Implementation: the KakeyaTurbo codec

3.1 Pipeline overview

Given a block $X \in \mathbb{R}^{n\times D}$ of KV vectors and weights $w \in \mathbb{R}^n_{\ge 0}$:

1. Compute the weighted mean $\mu = \sum_i w_i x_i / \sum_i w_i$.

2. Fit a weighted skeleton $U \in \mathbb{R}^{D \times d_{\text{eff}}}$ (§3.2).
3. Project each x_i into coefficient space $c_i = U^\top (x_i - \mu) \in \mathbb{R}^{d_{\text{eff}}}$.
4. Cluster $\{c_i\}$ via weighted spherical K-means with K centres.
5. Compute residuals $r_i = c_i - t_i c_{\text{seg}(i)}$ where t_i is the scalar projection onto the assigned cluster centre.
6. Zero-pad r_i to the nearest power of two, apply a Walsh–Hadamard rotation, and scalar-quantize with Lloyd–Max at b bits.
7. Store $(\mu, U, \{\text{centres}\})$ as the block's *skeleton* (bf16) and per-vector codes (seg_id, t_i, packed residual).

Decoding runs the pipeline in reverse, all in $O(D \cdot d_{\text{eff}})$ per vector.

3.2 Skeleton construction via randomized SVD

The weighted PCA (2) requires an eigendecomposition of the $D \times D$ covariance matrix, costing $O(nD^2 + D^3)$ per block. For Gemma-4's $D = 512$ this is a real production cost. We replace the exact eigendecomposition with the Halko–Martinsson–Tropp randomized SVD [18]:

Algorithm 1 Weighted randomized SVD (Algorithm 4.3 of [18])

Input: $X \in \mathbb{R}^{n \times D}$, weights w, target rank k, oversample p, power iterations q
$\mu \leftarrow$ weighted mean of rows of X under w
$A \leftarrow \text{diag}(\sqrt{w})(X - \mathbf{1}\mu^\top) \in \mathbb{R}^{n \times D}$
Draw $\Omega \in \mathbb{R}^{n \times (k+p)}$ with i.i.d. $\mathcal{N}(0, 1)$ entries
$Z \leftarrow A^\top \Omega \in \mathbb{R}^{D \times (k+p)}$
for $i = 1, \ldots, q$ **do**
    $Z \leftarrow A^\top A Z$ ▷ subspace power iteration
end for
$Q \leftarrow$ orthonormal basis of Z via QR decomposition
$B \leftarrow AQ \in \mathbb{R}^{n \times (k+p)}$
Compute thin SVD: $B = \hat{U} \Sigma \hat{V}^\top$
Return $U = Q\hat{V}$ (top k columns), singular values Σ

Proposition 3.1 (HMT, Theorem 10.6)**.** *For any target rank k, oversample $p \geq 2$, and power iterations q, the output U of Algorithm 1 satisfies*

$$\mathbb{E} \left\| A - QQ^\top A \right\|_2 \leq \left(1 + \frac{4\sqrt{k+p}}{p-1} \right)^{1/(2q+1)} \sigma_{k+1}(A), \tag{7}$$

where σ_{k+1} is the $(k+1)$-th singular value of A.

The rank cap is the point. A critical observation, validated empirically in §5, is that Algorithm 1 is *not* merely a cheap substitute for exact PCA. The target-rank parameter k becomes a *hard rank cap*: regardless of how long the data's true spectrum is, $d_{\text{eff}} \leq k$. Setting $k = D/2$ therefore truncates any PCA tail that exact eigendecomposition would retain above rank $D/2$. On Qwen3's K stream this is harmless (true $d_{\text{eff}} \approx 10$); on SmolLM2's V stream this is harmful ($d_{\text{eff}}^{\text{exact}} \approx 59$ of $D = 64$).

Complexity. Algorithm 1 costs $O(nDr)$ with $r = k + p$, versus $O(nD^2 + D^3)$ for exact weighted PCA. For Gemma-4 ($D = 512$, $n = 512$, $r = 72$) this is $\sim 40\times$ fewer operations per block.

3.3 RoPE-aware preprocessing of the K stream

Rotary position embeddings (RoPE) [19] rotate the key vector at position p by a position-dependent orthogonal matrix R_p: $k_{\text{post}} = R_p k_{\text{pre}}$. Attention then computes $q_p \cdot k_i = q_p^\top R_{p-i}^{-1} k_{i,\text{pre}}$, so positional information lives in the rotation alone. The post-RoPE key k_{post} is high-dimensional even when k_{pre} is low-rank.

Definition 3.1 (Inverse-RoPE preprocessor)**.** Given a block of post-RoPE keys $\{k_{\text{post},i}\}$ and their token positions $\{p_i\}$, the preprocessor computes $k_{\text{pre},i} = R_{p_i}^{-1} k_{\text{post},i}$ before feeding into the skeleton fit.

On the Qwen family (halfsplit-RoPE with learned W_k optimized for post-RoPE attention), we observe empirically (§5) that $\{k_{\text{pre},i}\}$ has a dramatically lower effective rank than $\{k_{\text{post},i}\}$. Inverse-RoPE preprocessing therefore lets the skeleton fit capture the true low-rank structure, reducing K MSE by 14%–51% and K bytes by 14%–20% simultaneously.

3.4 Residual turbo quantization

After skeleton subtraction and K-means re-centering, the residual $r_i \in \mathbb{R}^{d_{\text{eff}}}$ is compressed by a Gaussianisation + scalar-quantisation pipeline: zero-pad to next_pow_two(d_{eff}), apply a random-sign-flip Walsh–Hadamard rotation, and quantise each coordinate with a Lloyd–Max codebook of 2^b levels optimised for the unit Gaussian. We call this the *turbo* stage of KakeyaTurbo because it implements the residual-coding half of the rate–distortion sandwich in the Gaussian regime.

Proposition 3.2. *The WHT is isometric:* $\|\text{WHT}(r)\|_2 = \|r\|_2$*. For any zero-mean Gaussian* $r \sim \mathcal{N}(0, \sigma^2 I_d)$*, the rotated coordinates* $(\text{WHT}(r))_j$ *are i.i.d.* $\mathcal{N}(0, \sigma^2)$*.*

This is the rate–distortion optimality argument for scalar quantisation in the Gaussian regime. In KakeyaTurbo the residual r_i is not exactly Gaussian but is approximately so after the PCA + K-means subtractions; empirically the Lloyd–Max codebook matches the residual spectrum to within 5% on all seven benchmark models.

3.5 Design contract

The Rust implementation enforces the following constraints at compile time:

- No `unsafe` code anywhere in the crate (`unsafe_code = "forbid"`).
- No dynamic dispatch: the distortion function ρ is a compile-time type parameter, enabling monomorphisation and LLVM auto-vectorization of the inner loops.
- All skeleton tensors are stored in bf16; the hot path stays in f32.
- Every runtime parameter (b, r, variance_ratio, block_size, share_basis) is exposed through a single `CodecParams` struct and surfaced on every CLI.

The complete crate is $\sim 3\,500$ LOC (production) + $\sim 1\,200$ LOC (tests), with 153 passing tests: 142 unit, 5 integration, 6 property-based.

4 Experimental methodology

4.1 Models and environment

We benchmark on seven open-source decoder-only transformer models spanning four architectural families (Table 2).

Table 2: Benchmark corpus.

Model	Params	L	n_{kv} / hd	Attention family
Qwen2.5-0.5B-Instruct	0.5 B	24	2 / 64	GQA, halfsplit RoPE
Qwen3-0.6B	0.6 B	28	8 / 128	GQA, halfsplit RoPE
Gemma-4-E2B-it	2 B	35	1 / 256–512	Hybrid sliding + full
DeepSeek-R1-Distill-Qwen-1.5B	1.5 B	28	2 / 128	GQA, halfsplit RoPE
GLM-Edge-1.5B-chat	1.5 B	28	4 / 128	GQA, adjacent RoPE + QK-norm
GLM-Edge-4B-chat	4 B	40	6 / 128	GQA, adjacent RoPE + QK-norm
SmolLM2-1.7B-Instruct	1.7 B	24	32 / 64	MHA, halfsplit RoPE

All runs execute a real Hugging Face forward pass in bf16 with `attn_implementation="eager"` on a CPU-only 15 GB-RAM VM, using an identical long-form prompt (a $\sim$ 4 000-token text-continuation seed). KV tensors are captured from `DynamicCache` and fed to the Rust codec binary as KKTV-format files. No mock, no fallback, no simplification.

4.2 Baselines

Two baselines are compared against:

- **bf16 baseline**: the uncompressed cache at the same forward-pass configuration; 1.0× by definition.
- **TurboQuant turbo3** [4]: the reference Python implementation at 3 bits/coord on both K and V, running on the same captured tensors. Its per-vector cost is $3d/8 + 4$ bytes (K, fp32 norm) and $3d/8$ bytes (V).

4.3 Quality metric

For each codec configuration and each layer we compute the mean per-block reconstruction MSE on both K and V streams, excluding layer 0 (RoPE-degenerate). We report MSE *inflation* against the v1.2 baseline (exact PCA, $b = 3$) and adopt the same acceptance thresholds used in every ablation of the project:

MSE inflation	Verdict
$\leq 1.10\times$	ACCEPT (safe to ship)
$1.10\times < \cdot \leq 1.30\times$	MARGINAL (ship only if byte gain is large)
$> 1.30\times$	REJECT

4.4 Extrapolation

Ratios at context lengths beyond the measured 4 096 are computed by a byte-exact extrapolation script (`benchmarks/extrapolate_v1_2_b2_vs_turbo3.py`). For the KakeyaTurbo codec, only the V-side shared PCA basis is context-invariant; everything else scales linearly with context. For TurboQuant turbo3, all bytes scale linearly. For sliding-window layers both codecs pass through.

5 Benchmark results

5.1 Main result: tier-1 compression ratios

Table 3 reports the KakeyaTurbo tier-1 configuration ($b = 2$, randomized-SVD with $r = D/2$, oversample $p = 8$, power iterations $q = 2$), measured at ctx=4 096 and byte-exactly extrapolated to ctx=128k, against the bf16 baseline.

Table 3: KakeyaTurbo tier-1 compression ratios versus the bf16 baseline. MSE inflation is versus the exact-PCA $b = 3$ baseline of the same codec; the 0.42× K-MSE entry on Gemma-4 is a genuine Lloyd–Max RDO improvement (§5.4).

Model	ctx = 4 096	ctx = 128 k	K MSE infl.	V MSE infl.
Qwen2.5-0.5B	5.40×	5.46×	1.13×	1.15×
Qwen3-0.6B	**6.61**×	**6.65**×	1.08×	1.22×
Gemma-4-E2B (FA only)	**6.32**×	**6.72**×	0.42×	1.08×
DeepSeek-R1-Distill	5.98×	6.11×	1.13×	1.07×
GLM-Edge-1.5B	5.85×	5.91×	1.11×	1.13×
SmolLM2-1.7B	5.37×	5.37×	1.16×	1.61×
GLM-Edge-4B	5.83×	5.88×	1.12×	1.18×
Average	**5.91**×	**6.01**×		

Six of seven models achieve ACCEPT-level quality (MSE inflation $\leq 1.30\times$) at a compression ratio between 5.4× and 6.7× vs. the bf16 baseline. The lone outlier — SmolLM2, whose V stream has an essentially flat PCA spectrum — is analysed in §5.3. Head-to-head comparison against existing post-hoc KV compression algorithms is deferred to §5.7.

5.2 Tier-1.5: adding the inverse-RoPE preprocessor

For models using halfsplit RoPE without QK-norm (Qwen2.5, Qwen3, DeepSeek-R1-Distill), the inverse-RoPE preprocessor of §3.3 is applied to the K stream before the skeleton fit. Table 4 reports the measured K-stream improvements.

Table 4: Tier-1.5 inverse-RoPE preprocessor on the K stream: pre-RoPE vs post-RoPE compression. The "predicted total ratio" combines the improved K side with the unchanged V side via byte-exact recomposition.

Model	K MSE ratio (pre / post)	K bytes ratio (pre / post)	Predicted total ratio	Verdict
Qwen2.5-0.5B	0.49×	0.80×	$\sim 6.0\times$	ACCEPT
Qwen3-0.6B	0.86×	0.86×	$\sim 7.1\times$	ACCEPT
DeepSeek-R1-Distill	0.58×	0.81×	$\sim 6.8\times$	ACCEPT
Gemma-4-E2B	0.95×	1.42×	(worse)	REJECT
GLM-Edge-1.5B	1.13×	1.03×	(worse)	REJECT
GLM-Edge-4B	1.12×	1.03×	(worse)	REJECT
SmolLM2-1.7B	0.92×	0.96×	marginal	MARGINAL

The clean architectural bifurcation confirms our hypothesis: the inverse-RoPE preprocessor is useful exactly where RoPE is applied in its canonical halfsplit form without QK-norm, and harmful on architectures that use a different rotation (Gemma-4) or a post-QK-norm key projection (GLM-Edge).

5.3 The SmolLM2 outlier: a structural limit

SmolLM2 is the only benchmark model where the (b, r) Pareto frontier has an obstruction: no configuration simultaneously achieves the full $\geq 5\times$ tier-1 compression *and* stays within the ACCEPT $\leq 1.10\times$ MSE band. Table 5 enumerates the frontier.

Table 5: SmolLM2-1.7B: exhaustive knob search on the (b, r) frontier. No configuration achieves both the full tier-1 ratio ($\geq 5\times$) and ACCEPT V-MSE ($\leq 1.10\times$).

Configuration	Ratio	V MSE infl.	Verdict
v1.2 $b = 3$ exact (reference)	3.09×	1.00×	ACCEPT (baseline)
$b = 2$ sym $r = 32$	5.37×	1.61×	REJECT (V)
b_K=2, b_V=3, r=32	4.98×	1.54×	REJECT (V)
$b = 2$, r_K=32, r_V=64	4.47×	1.13×	MARGINAL
b_K=2, b_V=3, r_V=64	3.94×	1.00×	ACCEPT (below tier-1 ratio)

The root cause is structural: SmolLM2 is an MHA model with head dimension 64, and its V-stream PCA spectrum is essentially flat; exact PCA requires $d_{\text{eff}} = 59$ of $D = 64$ to capture 95% variance. There is no rank tail to truncate. Any $r < D$ pays the truncated eigenvalues in full MSE, while $r = D$ loses the compression advantage. This Pareto gap is intrinsic to the architecture, not a codec defect. We document SmolLM2 as a *tier-2 capability* in the product surface.

5.4 MSE measurements across configurations, models, and codec families

Reconstruction MSE is the primary quality metric throughout this paper. This subsection reports the full MSE picture of KakeyaTurbo along two axes: (i) codec configuration ($b = 3$ exact PCA vs. $b = 2$ exact PCA vs. $b = 2$ randomized SVD), and (ii) stream (K vs. V). Head-to-head comparison against existing compressors is deferred to §5.7. All values are mean per-block reconstruction MSE on real captured tensors, averaged across all full-attention layers excluding layer 0 (RoPE-degenerate).

Configuration sweep (K stream). Table 6 reports the transition from the v1.2 baseline ($b = 3$ exact PCA) through the bit-width reduction ($b = 2$ exact) to the final v1.3 tier-1 configuration ($b = 2$ randomized PCA with $r = D/2$).

Table 6: K-stream MSE across the three codec configurations at ctx=4096.

Model	$b = 3$ exact	$b = 2$ exact	$b = 2$ rsvd	$\frac{b=2e}{b=3e}$	$\frac{b=2r}{b=3e}$
Qwen2.5-0.5B	$9.89{\cdot}10^{-1}$	$1.10{\cdot}10^{0}$	$1.11{\cdot}10^{0}$	1.11×	1.13×
Qwen3-0.6B	$1.35{\cdot}10^{0}$	$1.44{\cdot}10^{0}$	$1.46{\cdot}10^{0}$	1.07×	1.08×
Gemma-4-E2B	$3.78{\cdot}10^{-3}$	$1.58{\cdot}10^{-3}$	$1.58{\cdot}10^{-3}$	0.42×	0.42×
DeepSeek-R1-Distill	$5.26{\cdot}10^{-1}$	$5.83{\cdot}10^{-1}$	$5.93{\cdot}10^{-1}$	1.11×	1.13×
GLM-Edge-1.5B	$7.65{\cdot}10^{-1}$	$8.35{\cdot}10^{-1}$	$8.49{\cdot}10^{-1}$	1.09×	1.11×
SmolLM2-1.7B	$6.73{\cdot}10^{-1}$	$7.63{\cdot}10^{-1}$	$7.80{\cdot}10^{-1}$	1.13×	1.16×
GLM-Edge-4B	$6.89{\cdot}10^{-1}$	$7.56{\cdot}10^{-1}$	$7.70{\cdot}10^{-1}$	1.10×	1.12×

Two observations stand out. First, the transition from exact PCA to randomized PCA at $r = D/2$ adds only 1%–2% additional K-MSE on top of the $b = 3 \rightarrow b = 2$ bit-width change — the randomized skeleton is essentially exact on the K stream because K is already low-rank (top-10 of $D = 128$ captures 95% variance on Qwen3 / DeepSeek). Second, Gemma-4 K-MSE is an outlier: $b = 2$ *improves* K quality relative to $b = 3$ (0.42×), because the $b = 3$ Lloyd-Max codebook over-quantizes the near-zero residual on Gemma's high-head-dim K. This is a genuine RDO win, not a measurement artefact, and it is the only ACCEPT-without-tradeoff result in the matrix.

Configuration sweep (V stream). Table 7 gives the corresponding V-stream MSE trajectory. V is generally more tolerant than K under MSE distortion, except in SmolLM2's MHA

head_dim=64 regime (see §5.3).

Table 7: V-stream MSE across the three codec configurations at ctx=4096.

Model	$b = 3$ exact	$b = 2$ exact	$b = 2$ rsvd	$\frac{b=2e}{b=3e}$	$\frac{b=2r}{b=3e}$
Qwen2.5-0.5B	$2.33 \cdot 10^{-1}$	$2.64 \cdot 10^{-1}$	$2.68 \cdot 10^{-1}$	1.13×	1.15×
Qwen3-0.6B	$4.34 \cdot 10^{0}$	$4.73 \cdot 10^{0}$	$5.29 \cdot 10^{0}$	1.09×	1.22×
Gemma-4-E2B	$2.81 \cdot 10^{-1}$	$3.07 \cdot 10^{-1}$	$3.03 \cdot 10^{-1}$	1.09×	1.08×
DeepSeek-R1-Distill	$4.82 \cdot 10^{-1}$	$5.26 \cdot 10^{-1}$	$5.15 \cdot 10^{-1}$	1.09×	1.07×
GLM-Edge-1.5B	$3.64 \cdot 10^{-1}$	$3.96 \cdot 10^{-1}$	$4.10 \cdot 10^{-1}$	1.09×	1.13×
SmolLM2-1.7B	$2.68 \cdot 10^{-1}$	$3.03 \cdot 10^{-1}$	$\mathbf{4.33 \cdot 10^{-1}}$	1.13×	**1.61**×
GLM-Edge-4B	$4.17 \cdot 10^{-1}$	$4.55 \cdot 10^{-1}$	$4.93 \cdot 10^{-1}$	1.09×	1.18×

The SmolLM2 V-MSE inflation of 1.61× is the REJECT signal that creates the tier-2 fallback documented in §5.3. On all other models V-MSE inflation stays $\leq 1.22\times$, within the MARGINAL band.

Inverse-RoPE K-MSE (tier-1.5). Table 8 reports the full per-model pre/post-RoPE K-MSE numbers underlying the tier-1.5 decision. Layer 0 is excluded (RoPE is identity at position 0).

Table 8: K-stream MSE under the inverse-RoPE preprocessor versus post-RoPE (ctx=4096). ↓ means MSE decreased; ↑ means it increased.

Model	post-RoPE K-MSE	pre-RoPE K-MSE	ratio	Verdict
Qwen2.5-0.5B	$1.33 \cdot 10^{0}$	$6.50 \cdot 10^{-1}$	0.49× ↓	ACCEPT
Qwen3-0.6B	$2.36 \cdot 10^{0}$	$2.03 \cdot 10^{0}$	0.86× ↓	ACCEPT
DeepSeek-R1-Distill	$6.53 \cdot 10^{-1}$	$3.76 \cdot 10^{-1}$	0.58× ↓	ACCEPT
SmolLM2-1.7B	$1.63 \cdot 10^{0}$	$1.50 \cdot 10^{0}$	0.92× ↓	MARGINAL
Gemma-4-E2B	$1.58 \cdot 10^{-3}$	$1.51 \cdot 10^{-3}$	0.95× ↓	REJECT (bytes worsen)
GLM-Edge-1.5B	$9.52 \cdot 10^{-1}$	$1.07 \cdot 10^{0}$	1.13× ↑	REJECT
GLM-Edge-4B	$9.00 \cdot 10^{-1}$	$1.01 \cdot 10^{0}$	1.12× ↑	REJECT

The clean bifurcation (halfsplit-RoPE-without-QK-norm models accept; all others reject) is the foundation of the tier-1.5 capability table in §5.

Per-layer K-MSE distribution. Beyond the aggregate mean, the per-layer MSE distribution is informative because attention-critical layers may sit at distribution tails. Table 9 gives the Qwen3-0.6B K-stream quantiles under tier-1.

Table 9: Per-layer K-MSE distribution on Qwen3-0.6B under tier-1, over 27 full-attention layers (L0 excluded).

Statistic	K-MSE
min (layer 13)	$8.08 \cdot 10^{-1}$
25th percentile	$9.70 \cdot 10^{-1}$
median	$1.14 \cdot 10^{0}$
75th percentile	$1.89 \cdot 10^{0}$
max (layer 4)	$3.15 \cdot 10^{0}$
mean	$1.46 \cdot 10^{0}$

The max-to-min spread across layers is 3.9×. Because each layer fits its own low-rank

skeleton (the RSVD is per block, not global), worst-case layers do not diverge further than this spread — a property we will contrast with data-oblivious codecs in §5.7.

5.5 Pareto summary across configurations

Ranking KakeyaTurbo configurations on the (ratio, quality) plane:

- Exact PCA with $b = 3$: lowest ratio ($\sim 3\times$), highest quality (ACCEPT by construction).
- Tier-1 ($b = 2$ exact): mid ratio ($\sim 4.5\times$), MARGINAL quality.
- Tier-1 ($b = 2$, $r = D/2$ randomized SVD): high ratio (5.4×–6.7×), MARGINAL quality on six of seven models, ACCEPT on Gemma-4.
- Tier-1.5 (tier-1 + inverse-RoPE): moves the Qwen / DeepSeek family further by +8% to +9% ratio at no quality cost.
- SmolLM2 is the only model where the full tier-1 ratio cannot be reached within the ACCEPT MSE band (see §5.3).

5.6 Flagship projections at 2026-Q2 scale

None of the April-2026 flagship open-source models (Qwen3-235B-A22B, DeepSeek-V3.1, Kimi-K2-Instruct, GLM-4.6, MiniMax-M2) is loadable on our 15 GB-RAM VM; their bf16 weights alone are 458–2000 GB. We therefore report byte-exact analytical extrapolation from architecturally-identical small-sibling proxies that *are* measured (Table 10).

Table 10: Byte-exact extrapolation to 2026-Q2 flagship open-source models. Per-vector bytes come from the measured small-sibling proxy; L and $n_{kv} \cdot \mathrm{hd}$ come from the flagship config.

Flagship	Params	L	hd	tier-1	tier-1.5
Qwen3-235B-A22B	235 B	94	128	6.53×	**7.13**×
DeepSeek-V3.1 (decomp)	671 B	61	192	5.92×	6.76×
Kimi-K2-Instruct (decomp)	1,000 B	61	192	5.92×	6.76×
GLM-4.6	355 B	92	128	5.84×	—
MiniMax-M2	229 B	62	128	5.92×	6.76×

DeepSeek-V3.1 and Kimi-K2 use Multi-Latent Attention (MLA); the reported ratios are on the decompressed K/V (what the attention computation sees). The additional gain from applying KakeyaTurbo to the stored MLA latent is an open question.

5.7 Head-to-head comparison with TurboQuant turbo3

This subsection consolidates all direct comparisons against the strongest existing post-hoc KV compression baseline, TurboQuant turbo3 [4], on the exact same captured tensors. We keep the comparison in one place so the rest of the paper can focus on KakeyaTurbo's own algorithm and RD boundary.

Compression ratio (tier-1, ctx=4 096 and ctx=128 k). Table 11 gives the ratio comparison on the seven benchmark models.

Table 11: Compression ratio vs. bf16 baseline: KakeyaTurbo tier-1 ($b = 2$, $r = D/2$) against TurboQuant turbo3 (3 bits/coord on K+V + fp32 K-norm), measured at ctx=4 096 and byte-exactly extrapolated to ctx=128 k.

Model	ctx = 4 096		ctx = 128k		Δ
	Turbo	turbo3	Turbo	turbo3	
Qwen2.5-0.5B	5.40×	4.92×	5.46×	4.92×	+11%
Qwen3-0.6B	**6.61**×	5.12×	**6.65**×	5.12×	+**30**%
Gemma-4-E2B (FA only)	**6.32**×	5.28×	**6.72**×	5.28×	+**27**%
DeepSeek-R1-Distill	5.98×	5.12×	6.11×	5.12×	+19%
GLM-Edge-1.5B	5.85×	5.12×	5.91×	5.12×	+15%
SmolLM2-1.7B	5.37×	4.92×	5.37×	4.92×	+9%
GLM-Edge-4B	5.83×	5.12×	5.88×	5.12×	+15%
Average	**5.91**×	5.08×	**6.01**×	5.08×	**+18%**

Six of seven models achieve a net positive ratio margin over turbo3 at ACCEPT-level quality; SmolLM2 reaches tier-1 ratio only at MARGINAL V-MSE (§5.3). Turning on the tier-1.5 inverse-RoPE preprocessor takes the Qwen / DeepSeek family further to +30% to +39% over turbo3 (cross-referencing Table 4).

Reconstruction MSE. Per-layer mean K-stream MSE on the identical captured tensors, summarised in Table 12, reveals that the ratio advantage is not traded for quality — on the Qwen / DeepSeek family the gap is large enough that KakeyaTurbo is strictly dominant.

Table 12: Head-to-head K-stream MSE on identical captured tensors (ctx=4 096). Ratio > 1 means KakeyaTurbo has lower MSE. turbo3 MSE numbers come from our prior comparison [5].

Model	Turbo K-MSE	turbo3 K-MSE	turbo3 / Turbo	Interpretation
Qwen2.5-0.5B	$1.11 \cdot 10^{0}$	$2.07 \cdot 10^{2}$	**186**×	KakeyaTurbo much lower
Qwen3-0.6B	$1.46 \cdot 10^{0}$	$8.99 \cdot 10^{1}$	**62**×	KakeyaTurbo much lower
DeepSeek-R1-Distill	$5.93 \cdot 10^{-1}$	$1.44 \cdot 10^{3}$	**2428**×	KakeyaTurbo much lower
Gemma-4-E2B	$1.58 \cdot 10^{-3}$	$2.96 \cdot 10^{-3}$	1.87×	KakeyaTurbo lower
GLM-Edge-1.5B	$8.49 \cdot 10^{-1}$	$2.81 \cdot 10^{-1}$	0.33×	turbo3 lower
GLM-Edge-4B	$7.70 \cdot 10^{-1}$	$1.48 \cdot 10^{-1}$	0.19×	turbo3 lower

On the Qwen / DeepSeek family (halfsplit RoPE without QK-norm) KakeyaTurbo's K-MSE is 62× to 2428× lower than turbo3, consistent with the asymmetric-rescue configurations the TurboQuant documentation recommends for these architectures. On GLM-Edge (adjacent-pairs RoPE with QK-norm) turbo3 has lower K-MSE because QK-norm already norm-stabilises the K representation and the Gaussianised scalar quantiser handles it cleanly. Gemma-4 sits in between, with KakeyaTurbo K-MSE 1.87× lower due to Gemma's high-head-dim low-rank K structure.

Cross-layer variance. A secondary but informative comparison is the spread of K-MSE across layers. KakeyaTurbo's per-block PCA re-fits each layer's low-rank structure, so the worst-case layer does not diverge from the best by more than 3.9× on Qwen3-0.6B (Table 9). turbo3's fixed rotation produces a $\sim 200\times$ cross-layer spread on the same tensor set, reflecting the fact that a data-oblivious rotation cannot adapt to per-layer anisotropy.

Summary. At comparable byte budget and same captured-tensor quality metric, KakeyaTurbo tier-1 strictly dominates turbo3 on six of seven benchmark models; tier-1.5 dominates on Qwen / DeepSeek by a further +30% to +39% in ratio; on GLM-Edge, turbo3 has an MSE advantage that reflects its natural fit with QK-norm architectures. Flagship extrapolation (Table 10) projects this ordering to 128k context on Qwen3-235B, DeepSeek-V3.1, Kimi-K2, GLM-4.6 and MiniMax-M2.

6 Related work

KV cache compression. H_2O [1] and Scissorhands [2] propose token eviction based on attention-score heavy hitters. StreamingLLM [20] observes the attention-sink phenomenon and preserves the first few tokens exact. KIVI [21] is a per-channel quantization scheme for K (asymmetric quantization per channel) and per-token for V. TurboQuant [4] is a data-oblivious WHT + Lloyd–Max scalar quantizer. We compare against turbo3 head-to-head on the same captured tensors in §5.

Low-rank KV methods. Kakeya-KV [5] is our prior work and the direct ancestor of the skeleton construction here. DMC [22] and LESS [23] learn low-rank projections during training. Our approach is post-hoc and training-free.

Randomized numerical linear algebra. The Halko–Martinsson–Tropp randomized SVD [18] is a textbook algorithm; we use it as a drop-in for exact PCA and observe that the implicit rank cap is the structural win. Frequent Directions [24] is a related streaming PCA sketch we defer to future work.

Kakeya geometry and harmonic analysis. The Kakeya problem originates in [6]. Its deep role in harmonic analysis began with Fefferman [7] (ball multiplier), continued through Bourgain's arithmetic-combinatorial bounds [8], Wolff's geometric hairbrush argument [9], and Katz–Łaba–Tao's $\mathbb{R}^3$ improvements [10]. The finite-field analogue was settled by Dvir's polynomial method [11]; Bennett–Carbery–Tao's multilinear Kakeya conjecture [12] and Guth's endpoint resolution [13] introduced the tools eventually used by Wang and Zahl [17] to resolve the Euclidean three-dimensional case. *Our use of this chain is formal, not intuitional*, as documented in §2.4: multilinear Kakeya is equivalent to Brascamp–Lieb [14, 15], which underpins Tropp's matrix concentration inequalities [16], which underpin the Halko–Martinsson–Tropp RSVD error bound [18]. The codec skeleton is therefore a literature-traceable shallow instance of the same toolchain used at its deep end in [17]. Proposition 2.2 makes the quantitative correspondence precise; the structural parallels (power iteration as multiscale induction, Gaussian probing as direction enumeration, singular-value distribution as Frostman measure) and the three rigorous disclaimers are in §2.4.

7 Limitations and future work

Quality measurement. We report reconstruction MSE, not perplexity or downstream-task accuracy. A full PPL sweep across LongBench/RULER is deferred to future work; early checks on Gemma-4-E2B at 2k context show greedy-decode token-level agreement with the bf16 baseline for the first 12 tokens. End-to-end quality at 128k context on flagship models remains unverified.

Flagship measurements. All flagship numbers are analytical extrapolation from measured proxies. Validation on ≥ 500 GB-RAM hardware is pending.

MLA models. DeepSeek-V3.1 and Kimi-K2 use MLA; our reported numbers compress the decompressed K/V, not the stored latent. Applying the codec to the MLA latent is a v1.4 open item.

GLM-family inverse-RoPE. GLM-Edge uses adjacent-pairs RoPE + QK-norm; our halfs-plit inverse is incompatible. A GLM-correct inverse rotation is tracked as v1.3.1.

Attention-aware extensions. The `weights: &[f32]` parameter in our API is a reserved socket for attention-score-driven importance, but is not yet wired. The v1.4 roadmap (public in the repository) proposes closing this loop by piping H_2O-style prefill attention scores into the weighted PCA fit.

SmolLM2-class models. MHA with head dim 64 and flat V spectra defeats rank truncation; the Pareto gap observed on SmolLM2 will likely appear on any future model with similar structure. We ship SmolLM2 as a tier-2 capability that requires an explicit quality-versus-bytes trade-off at the deployment policy layer.

8 Conclusion

We presented KakeyaTurbo, a training-free, three-stage KV cache compression codec, and established its rate–distortion boundary.

On the *algorithm* side, the three stages — weighted randomized SVD with rank cap $r = D/2$, per-architecture inverse-RoPE preprocessing on the key stream, and 2-bit Lloyd–Max residual quantization after WHT — are the separable optimisers of three nested subproblems of a single rate–distortion objective, not a pipeline of independent heuristics. The codec is parameterised entirely by a compile-time distortion metric and a per-vector weight, and runs in the full LLM inference pipeline (prefill + token-by-token decode + paged attention) with a block-streaming mode whose amortised overhead is under 10 μs per layer per decode step on H100.

On the *RD boundary* side, the randomized-SVD skeleton sits at the shallow end of an explicit four-step dependency chain whose deep end is used by Wang and Zahl in their recent resolution of the three-dimensional Kakeya conjecture. The chain yields an unconditional upper bound (Proposition 2.2) and a conjectural lower bound (Proposition 2.1) that closes unconditionally at $D = 3$ via Wang–Zahl. The rank cap $r = D/2$ is thereby derived as a Pareto-optimal point on the RD curve.

The supporting benchmark shows that KakeyaTurbo achieves 5.4×–6.7× compression against the bf16 baseline on six of seven open-source models at ACCEPT quality, with analytical projections to 7.1× on flagship Qwen3-235B. The head-to-head comparison against the strongest prior compressor (§5.7) confirms the ordering. Open items are limited and specific: end-to-end perplexity validation, GLM-Edge inverse-RoPE (adjacent-pairs + QK-norm), MLA-latent compression, and the attention-aware extension; each has a concrete roadmap in the public repository.

Reproducibility. All code, unit tests, integration tests, property-based tests, real-measurement JSON, and ablation decision documents are released under Apache-2.0. Repository:

https://github.com/FluffyAIcode/LLM-KV--Cache-compress

The v1.3 release state lives on branch `cursor/v1-3-rsvd-rope-aware-12f5` (Pull Request #11); the v1.4 roadmap is in `reports/v1_3_rsvd_rope/V1_4_V1_5_ROADMAP.md`.

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A Complete benchmark protocol

All benchmarks in this paper are reproducible via the following one-line invocation after cloning the repository and downloading a model:

```
python3 benchmarks/kakeyaturbo_v1_2_real_bench.py \
    --model-path models/<model-dir> \
    --model-name <model-name> \
    --context-tokens 4096 \
    --bit-width 2 \
    --pca-method randomized \
    --out-dir reports/v1_3_rsvd_rope/bench/<model>_rsvd_half/ctx_4096
```

The inverse-RoPE preprocessor adds:

```
python3 benchmarks/rope_aware_k_poc.py \
    --model-dir <model-dir> --model-name <model-name> \
    --ctx 4096 --rope-pairing halfsplit \
    --out-dir reports/v1_3_rsvd_rope/rope_poc/<model-name>
```

Each run produces a JSON manifest per layer plus an aggregate `summary.json`. All manifests are committed to the repository under `reports/v1_3_rsvd_rope/`.

B Codec parameter defaults

The reference v1.3 tier-1 configuration is:

Parameter	Default value
`bit_width`	2
`variance_ratio`	0.95
`k` (K-means centres)	16
`block_size`	512
`kmeans_max_iter`	32
`share_basis` (V stream)	true
`share_basis` (K stream)	false

The randomized-SVD parameters are:

Parameter	Default value
`target_rank`	$D/2$
`oversample` (p)	8
`power_iters` (q)	2
`seed_offset`	`0x9E37_79B9_7F4A_7C15`

Tier-1.5 additionally applies the inverse-RoPE preprocessor to the K stream before the codec sees it, using the model's `rope_theta` and halfsplit pairing convention.

C Qualitative ablation trail

The v1.3 configuration is the endpoint of seven ablations run over the project's lifetime. For completeness, the full decision chain is:

1. **v1.0 → v1.2 A**: bf16 storage of skeleton (halves skeleton bytes, no quality loss). ACCEPT, shipped.
2. **v1.2 B'**: V-stream layer-pooled PCA basis. ACCEPT (V only), shipped. Rejected on K due to family-dependent divergence.
3. **K-side d_{eff} truncation** + outlier compensation: every 0.05 drop in `variance_ratio` doubles K MSE; outliers recover $\sim 25\%$. **REJECT**.
4. **Block size doubling** (512 → 1024): MSE inflation 1.14× on average with 16% byte savings. **MARGINAL**, not shipped as default.
5. **Cross-layer basis sharing** on K: 2×–15× MSE inflation on Qwen/DeepSeek family, ACCEPT on Gemma/GLM-Edge. **REJECT as default, accept as per-family knob.**
6. **Bit width sweep** ($b = 3 \to b = 2$): +4% to +23% ratio, MSE inflation 1.07×–1.13×. **ACCEPT, new default.**
7. **Randomized SVD with rank cap** $r = D/2$: combined with $b = 2$, pushes tier-1 compression ratio to 5.4×–6.7× on 6/7 models at ACCEPT quality. **ACCEPT, new default.**
8. **Inverse-RoPE K preprocessor**: 14%–51% K MSE drop on halfsplit-RoPE family, no-op or harmful elsewhere. **ACCEPT as per-family knob.**

The full narrative is in the repository's `reports/v1_3_rsvd_rope/DECISION.md`.